

LIQUID SELECTION SYSTEM FOR *BASMATI* *BAR* TRANSFORMANTS

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Using pAHC25, a plasmid containing *gus* and *bar* gene coding regions driven by the maize ubiquitin promoter, first exon and first intron (*Ubi* 1), 5 day precultured mature embryos of basmati rice (*Oryza sativa* L.) cultivar Karnal Local were subjected to microprojectile bombardment. 1510 embryos were processed through a solid media selection system with 4mg/l phosphinothricin (PPT) supplemented modified MS medium (35mM KNO₃ and 5mM (NH₄)₂SO₄ instead of 19mM KNO₃ and 21mM NH₄NO₃ of MS basal medium) (Khanna and Raina in press). 1101 embryos were processed through a liquid selection system using modified MS medium supplemented with 30g/l sorbitol, 1g/l caesin acid hydrolysate, 2g/l proline, 500mg/l glutamine, 12.5 mM phosphate buffer (pH 5.8), 0.5 mg/l kinetin and 0.5 mg/l NAA.

After reducing the total nitrogen content of the media to half of its original level (17.5 mM KNO₃ and 2.5 mM (NH₄)₂SO₄) at the end of three weeks, somatic embryogenesis could be induced in the liquid selection medium, in otherwise necrotic looking calli. 20-50 somatic embryoids could be seen budding out from each primary calli. Most of them died within a week due to selection pressure.

The resistant ones grew in size, detached themselves from the parent callus, and produced adventitious embryos. These microcalli were regenerated on MS regeneration medium and 14 putative transformants were recovered using this approach. 12 of them rooted in PPT supplemented 1/2 MS medium. PCR and Southern analysis confirmed 10 of these to be transformants for *bar* gene. Co-transformation frequency of *gus* was more than 50%. The solid selection procedure failed to give any transformants and while most of the calli died during primary selection, a few proliferating structures recovered from the primary calli died during secondary selection.

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Expression of the *CpII* gene driven by the constitutively active promoter of the rice *act1* gene (*Act1/CpII*) leads to high level accumulation of the *CpII* protein in transgenic rice plants. (Cont. next page.)